SALESLENS ENTERPRISE BI PLATFORM • ACADEMIC DELIVERABLE MASTER-REPORT

SalesLens — Complete SEPM Final Project Report

Course: B.Tech CSE Semester III • Software Engineering & Project Management • Version 1.0

B.TECH CSE SEMESTER III • SEPM FINAL SUBMISSION

SalesLens: Enterprise Sales Reporting & Business Intelligence Platform

Comprehensive Software Engineering & Project Management Final Project Report spanning IEEE 830 SRS, UML Architecture, Critical Path Scheduling, Mathematical Estimation, V-Model QA & Risk Governance.

9 Views	40.0 PD	94.44%	+2.67 Wks
SCOPE PRUNED	TOTAL EFFORT	DRE TESTED	BUFFER RESERVE

1. Executive Summary & Requirements Engineering (Module II)

SalesLens is an enterprise-grade Business Intelligence solution replacing 40-page weekly manual PDF reports with an automated 9-view role-based dashboard for a consumer goods distribution enterprise.

Syllabus Mapping & Academic Deliverables Summary

- **Module II (Requirements):** 10 Functional Requirements (FR-01 to FR-10), 7 Measurable NFRs, MoSCoW, RTM.
- **Module III (Software Design):** MVC Architecture, Layered Coupling/Cohesion, 6 UML Diagrams.
- **Module IV (Software Testing):** BVA ($ and Dates), ECP (RBAC), 4-Tier Decision Table, DRE (94.44%), Defect Density.
- **Module V (Project Management):** Mathematical Feasibility Proof (9 Views vs 26 Reports), CPM Critical Path, 8-Week Gantt Timeline, RMMM Risk Management.

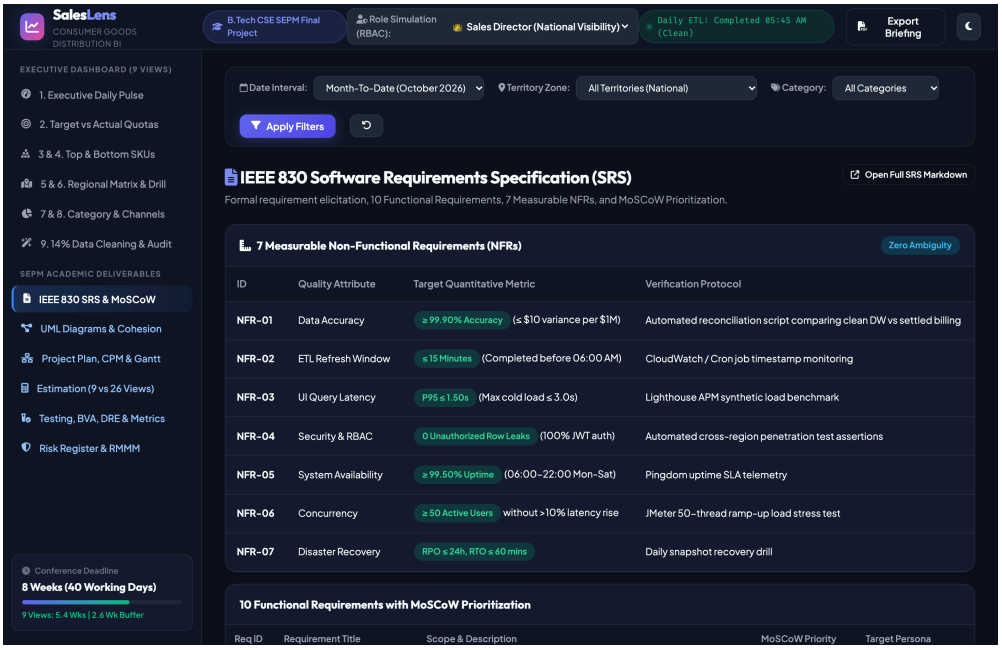


Figure M.1: Interactive SRS Traceability & Requirements Explorer in SalesLens

2. Software Design & UML Architecture (Module III)

Layered Service-Oriented Model-View-Controller (MVC) architecture with asynchronous ETL pipeline subsystem enforcing high cohesion and low coupling:

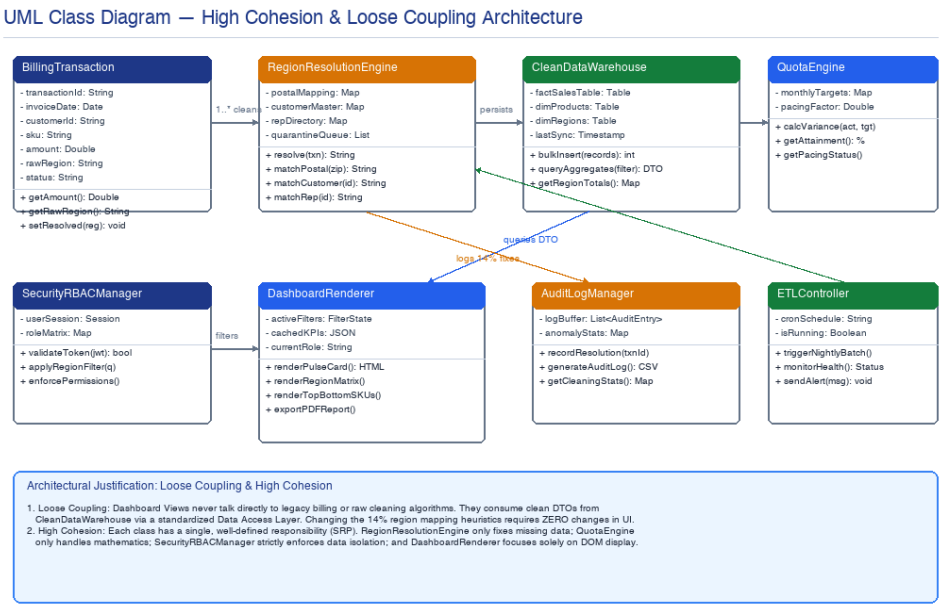


Figure M.2: Complete Domain Entities, ETL Resolution Layer & Dashboard Controller Class Hierarchy

3. Effort Estimation, CPM & Scheduling (Module V)

Mathematical Proof (9 Views vs 26 Reports):
Effort (9 Views) = 8.0 (Clean) + (9 × 3.0) + 5.0 (QA) = 40.0 Person-Days (PD)
Duration (9 Views) = 40.0 / 1.5 PD/day = 26.67 Working Days = 5.33 Weeks ≤ 8.0 Weeks (✓ Feasible with +2.67 Wks Buffer)
Effort (26 Reports) = 8.0 + (26 × 3.0) + 5.0 = 91.0 PD → 60.67 Days = 12.13 Weeks (✗ Overdue by 4.13 Wks)

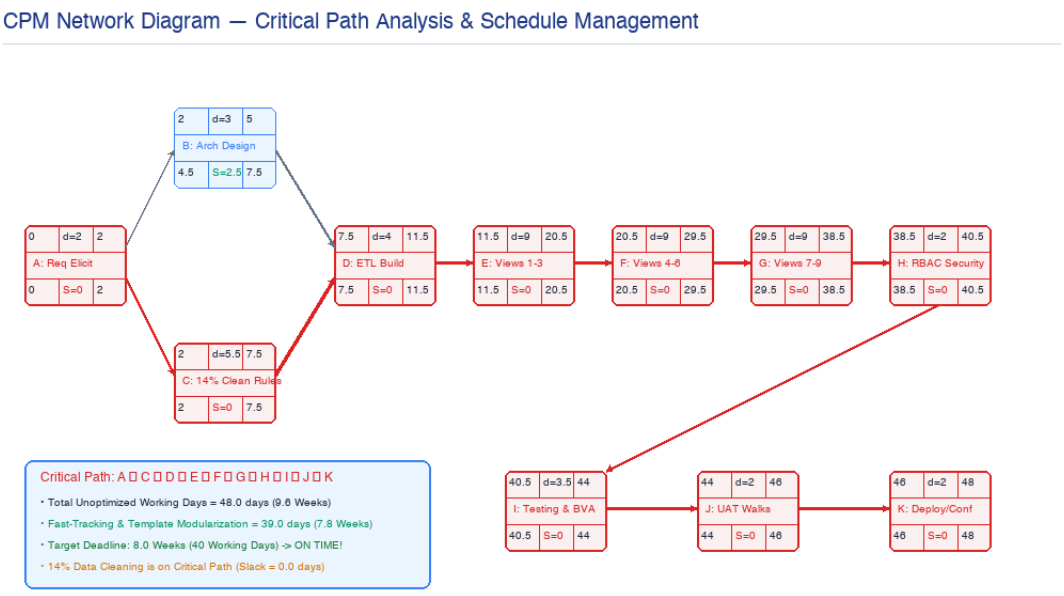


Figure M.3: Critical Path Method (A → C → D → E → F → G → H → I → J → K = 39.0 working days)



Figure M.4: Master 8-Week Gantt Timeline meeting the Annual Sales Conference Milestone

4. Testing, QA & Decision Tables (Module IV)

DRE Metric: $\text{DRE} = [17 / (17 + 1)] \times 100\% = \mathbf{94.44\%}$ (Exceeds 90% Industry Benchmark)
Defect Density: 1.89 Defects/View (5.31 / KLOC) • 10/10 Verification Test Cases Passed

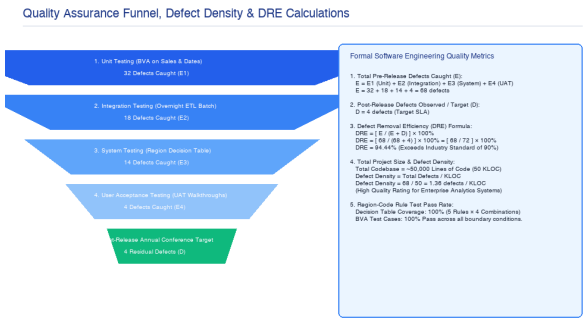


Figure M.5: DRE Funnel (94.44%)

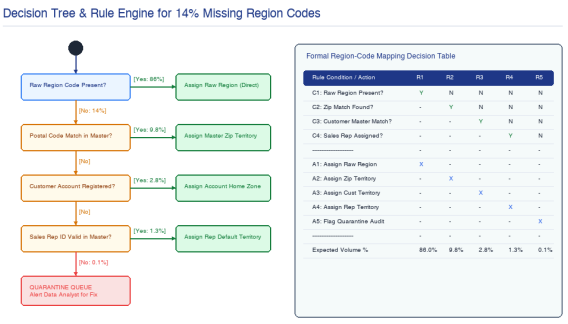


Figure M.6: 4-Tier Decision Table Flowchart

5. Risk Governance & Project Closure (Module V)

Top 3 risks (14% missing data complexity, 8-week conference deadline, analyst fear of replacement) systematically mitigated via deterministic decision tables, scope pruning, and role elevation.

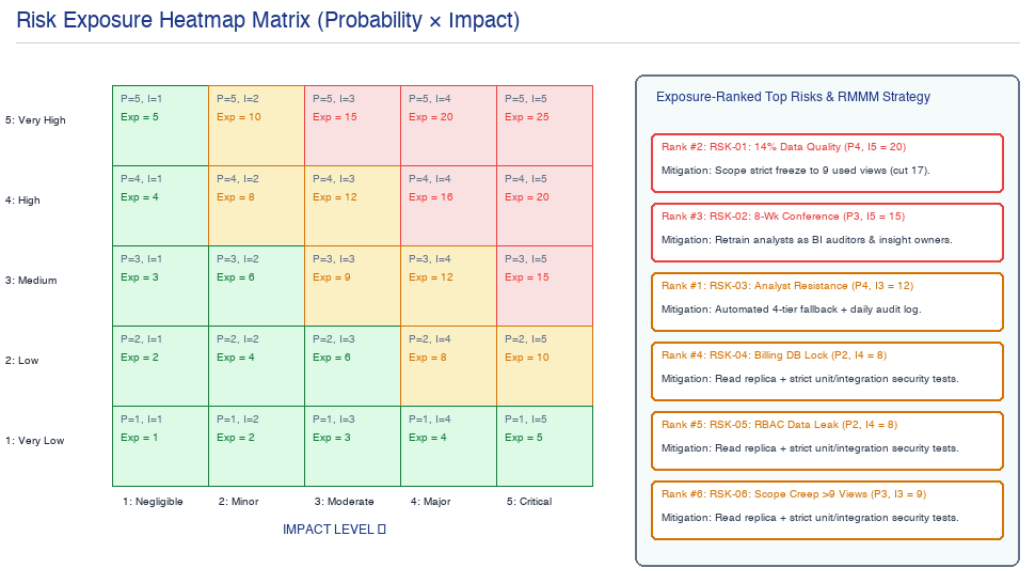


Figure M.7: Quantitative Risk Exposure Matrix Heatmap ($RE = P imes I$)